Yashas R

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EDUCATION

Jyothy Institute of Technology, Bangalore Sep 2022 – Jun 2026

B.E. in Computer Science and Engineering | CGPA: 8.62/10.0

Relevant Coursework: DSA, OOPS, DBMS, OS, Computer Networks, Machine Learning

WORK EXPERIENCE

Technical Intern, Super Mentr Technologies Pvt Ltd Feb 2026–May 2026

- Designed and developed Cognito, an autonomous AI research agent, implementing a ReAct reasoning loop with GPT-4o for iterative multi-query knowledge retrieval and synthesis of structured research reports.
- Built and debugged a Retrieval-Augmented Generation (RAG) pipeline using ChromaDB as a persistent vector database with OpenAI text-embedding-3-small, applying cosine similarity threshold filtering to improve retrieval precision.
- Engineered a modular software architecture separating document ingestion, web scraping (BeautifulSoup), vector storage, and multi-mode agent reasoning; executed test plans and iteratively debugged issues across components.
- Collaborated within a small engineering team through code reviews and integration testing to deliver production-ready features on schedule.

TECHNICAL PROJECTS

CodeVault – Full-Stack Code Snippet Manager | GitHub Apr 2026- May 2026

React), TailwindCSS, FastAPI, Node.js, PostgreSQL, Redis, JWT, GitHub Actions

- Architected a full-stack code snippet manager with JWT-based authentication, a RESTful FastAPI backend, and PostgreSQL persistence, implementing ownership-scoped CRUD operations validated via automated cross-user security tests.
- Designed a Redis-based cache versioning system that invalidates all cached queries per user in O(1) on writes, and replaced offset pagination with cursor-based pagination for efficient scaling on large tables.
- Implemented PostgreSQL full-text search using tsvector columns with GIN indexing and auto-update triggers, replacing non-indexable ILIKE queries, alongside sliding-window Redis rate limiting on authentication endpoints.
- Built a CI/CD pipeline with GitHub Actions running a 10-test pytest suite (auth, CRUD, caching, pagination, security) on every push, ensuring consistent code quality and deployment readiness.

Colorize – GAN Based Image Colorization | GitHub Dec 2025 – Feb 2026

Python, PyTorch, ResNet18, fastai, Streamlit, Fine Tuning, Transfer Learning

- Built a conditional PatchGAN with a custom U-Net generator and PatchDiscriminator, trained on 10,000 COCO images using a 3-stage progressive pipeline (baseline training, supervised pre-training, adversarial fine-tuning), reducing convergence time by 70%.
- Integrated a ResNet18 ImageNet backbone into the generator via transfer learning with fastai's DynamicUnet, enabling grayscale image colorization in the LAB color space, with staged checkpointing across all 3 training stages, deployed as a real-time Streamlit web app.

MailMind.io – AI Powered Email Drafter | GitHub Mar 2026 – Apr 2026

Python, LangChain, LangGraph, OpenAI API, Streamlit

- Built a tool-augmented LangGraph agent with a custom ReAct loop and conditional routing between 6 LangChain tools, orchestrating recipient, subject, and content generation with GPT-4o in a stateful multi-turn workflow.
- Deployed a Streamlit web app with real-time split-panel draft preview, dynamic system prompting with live draft state injection, and local draft persistence capable of autonomously drafting, refining, and saving professional emails.

TECHNICAL SKILLS

Programming: C, Python, JavaScript, SQL

Web: Streamlit, FastAPI, React.js, Node.js, Express.js, MongoDB, PostgreSQL, REST APIs

AI/ML: Numpy, Pandas, Matplotlib, Scikit-learn, PyTorch, TensorFlow, Keras, HuggingFace Transformers, LangChain, LangGraph, LLMs, Crew AI, Retrieval Augmented Generation (RAG), GANs

Developer Tools: Git, GitHub, GitHub Actions, VS Code, Colab, PyCharm, Postman, Redis, Docker

Problem Solving: 300+ problems solved across LeetCode, HackerRank, GeeksforGeeks

ACHIEVEMENTS & LEADERSHIP

- Vice President, Rotaract Club, JIT-3192 (2024–25): led 40+ members, volunteering and organizing community service projects including educational outreach, old age home visits, and District Conference representation.
- Co-Author: "IoT-Enabled Machine Learning for Detecting Nutrition Deficiency Disorders in Mulberry Leaves" — accepted in Cloud-Enabled Artificial Intelligence of Things (AIoT), Taylor & Francis – CRC Press, 2024.
- IBM Mindscape: Finalist – Led a team of 3 for the Theme Mental Health awareness among 300+ teams (2025).